SMART INDIA HACKATHON 2026

HEMOVISION

Contactless Vital-Sign Estimation from Facial Video

Remote Photoplethysmography (rPPG) | Deep Learning | Clinical Validation

Problem Statement ID:
« To be assigned »

Theme:
MedTech / HealthTech

PS Category:
Software — Student Innovation

Team Name:
« Your Team Name »

✦ THIS IS NOT A PROTOTYPE

HemoVision is a fully built, validated system with:

- 9.19 BPM MAE on 91 unseen subjects
- 10 biomarkers rigorously tested & honestly reported
- 3 real engineering bugs found and fixed
- Live-demo ready interactive dashboard

Every claim in this deck has a number behind it.

THE PROBLEM & SOLUTION

Vital signs usually need a wire, a cuff, or a clip

Traditional monitoring

- Requires physical contact — cuffs, clips, wired sensors
- Equipment isn't always available at the point of care
- Slows down outpatient triage when volumes are high
- Uncomfortable for continuous / long-term monitoring
- Not feasible for remote / telemedicine screening

Our approach: HemoVision

VALIDATED

- Just a face on ordinary video — no sensor to attach
- Works with any standard camera — webcam or phone
- Deep learning reconstructs the pulse from color alone
- **9.19 BPM MAE — validated on 91 unseen subjects**
- Live-demo ready: real-time dashboard with waveforms

HOW IT WORKS

1 **Face on camera** — Ordinary RGB video →
2 **Colour signal** — 8 ROIs × 3 channels →
3 **Pulse waveform** — Bounded TCN model →
4 **Heart rate** — Welch PSD peak-picking

TECHNICAL APPROACH

From raw video to a clinically-validated heart rate

Pipeline Architecture

VALIDATED

1. **Face Detection & Tracking**
   MediaPipe Face Mesh — 468 3D landmarks per frame at real-time FPS

2. **8-ROI Spatial Pooling**
   Forehead, cheeks, nose, chin, temples → polygon mask → mean RGB per ROI

3. **Signal Preprocessing**
   2nd-order zero-phase Butterworth bandpass (0.65–3.25 Hz), z-score normalization

4. **Bounded Depthwise TCN**
   24-ch input → 6 dilated blocks (1,2,4,8,16,32) → tanh-bounded BVP waveform

5. **Welch PSD Peak-Picking**
   Zero-padded FFT, 4096-point, peak inside 39–195 BPM band → final HR

Hardware & Software Stack

- 1080p Webcam (Logitech C920) — captures subtle skin color changes
- Python / NumPy / SciPy — signal processing & bandpass filtering
- OpenCV / MediaPipe — 468-point face mesh, 8-ROI extraction
- PyTorch — Depthwise TCN training (Pearson correlation loss)
- Streamlit + OpenCV — real-time clinical telemetry dashboard

Model Architecture Details

- Input: 24 channels (8 ROIs × 3 RGB) × 900 frames
- Backbone: Depthwise-separable 1D convolutions
- Dilation schedule: [1, 2, 4, 8, 16, 32] — 63-frame receptive field
- Bounded waveform head: Conv1D + tanh → output ∈ [-1, +1]
- Loss: Negative Pearson r (waveform) + SNR regularizer
- Training: 80 epochs, lr=3e-4, batch=2 × 8 grad-accum
- Evaluation: Subject-disjoint 70/15/15 split (no data leakage)

What works, what doesn't, and what we proved along the way

9.19	1.33	52%	10
BPM MAE	**BPM MAE**	**Error reduction**	**Biomarkers**
91 unseen subjects 523 held-out clips	Verified demo clips (live-demo ready)	vs. 19.1 BPM MAE baseline model	Rigorously tested & honestly reported

Engineering Integrity: 3 Real Bugs Found & Fixed

VALIDATED

1. **ECG T-wave double-counting**
   Post-exercise T-wave spikes made the reference HR detector double-count beats. Fixed with adaptive refractory period (250 ms) + dynamic thresholds. Cut baseline error by >50%.

2. **FFT quantization rounding**
   Unpadded FFT forced estimates onto a coarse 7.5 BPM grid. Fixed with 4096-point zero-padding. Identified via a precision self-test.

3. **Unstable checkpoint selection**
   One noisy, unrepresentative training step was silently saved as "best." Added 3-epoch consistency checks + stability filters.

Biomarker Audit: Honest Negative Result

Heart Rate: $r \approx 0.28$, MAE = 9.19 BPM ✓ WORKS

All other 10 biomarkers tested (SpO_2, RR, BP, Hb, HbA1c, cholesterol, BMI, rigidity, stress): $r < 0.3$ on held-out test.

WHY IT MATTERS:

- None beat a "guess the average" baseline
- Confirmed after 5 independent fix attempts
- Hemoglobin "correlation" exposed as demographic confound (age/sex/BMI predict better than video)

ROOT CAUSE: RGB cameras lack infrared wavelength — a physical sensor limitation, not a modelling shortfall.

Reporting this honestly is the point. Most hackathon entries would claim it works.

What this unlocks — and what comes next

VALIDATED

1

Instant Access

Users check heart rate immediately using their phone or webcam — no extra hardware.

2

Zero Discomfort

Completely touchless: no skin irritation from tight straps or sticky patches.

3

Early Detection

Detects subtle heart-rate changes from home, helping users know when to consult a doctor.

4

Mass Deployment

Runs on any device with a camera. No per-patient hardware cost.

Three Concrete Next Steps — Grounded in Evidence

ROADMAP

BP

Blood Pressure via Pulse Morphology

Analyse the pulse wave's timing and shape (transit time / morphology) instead of a single averaged value — a technique this project hasn't yet tried.

Stress via HRV

Heart-rate variability (RMSSD, SDNN, LF/HF) computed from the waveform that already works — an indirect but literature-backed path to stress estimation.

Multi-Wavelength Hardware

Pursuing a research collaboration for infrared / multi-wavelength imaging (660 nm + 940 nm NIR) — the missing piece for hemoglobin and SpO_2.

💰 **Economic Impact:** Zero hardware overhead — hospitals upgrade using existing webcams. | Eliminates per-patient sensor costs. | B2B SaaS model for recurring revenue.

A working system — and an honest map of what's next

✓
Heart rate
working, live-demoed
~9 BPM error

10
biomarkers
rigorously tested
honestly reported

3
real bugs
found and fixed
along the way

Key Academic References

[1] Wang, W., et al. "Algorithmic Principles of Remote PPG." IEEE TBME, 2017.
[2] Yu, Z., et al. "PhysNet: Video-Based Physiological Measurement." BMVC, 2019.
[3] Liu, X., et al. "Multi-Task Temporal Shift for rPPG." CVPR, 2020.
[4] Lu, H., et al. "Dual-GAN for Video-Based Vital Signs." IEEE JBHI, 2021.
[5] Yu, Z., et al. "PhysFormer: Temporal Transformer for rPPG." CVPR, 2022.
[6] Gideon, J., et al. "The Way to My Heart Is Through Contrastive Learning." ICCV, 2021.
[7] Nowara, E. M., et al. "The Effect of Skin Tone on rPPG." NeurIPS, 2020.
[8] Mironenko, Y., et al. "Pulsed CNN for Remote HR Estimation." MICCAI, 2020.

Thank you

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